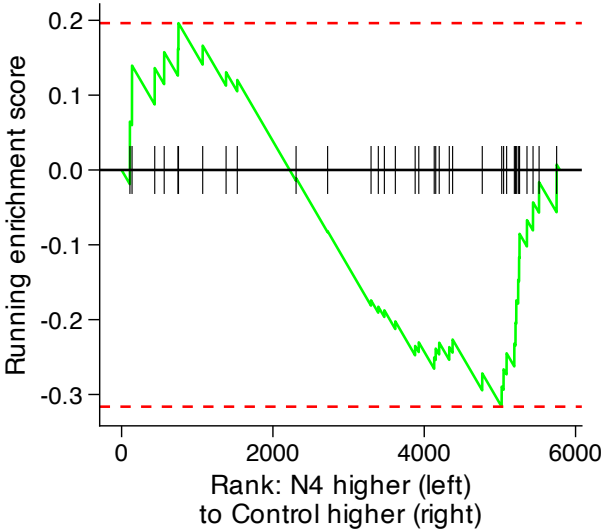
## Epithelial cluster 3 Hallmark: N4 minus Control

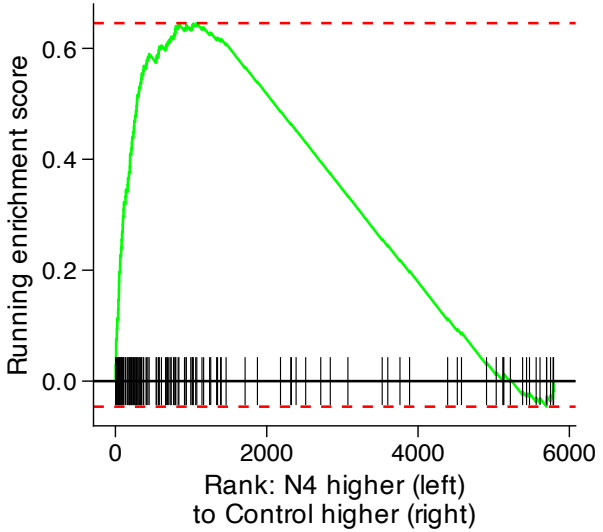
### HALLMARK BILE ACID METABOLISM

N4 minus Control | NES -1.18
BH FDR 0.31 | favours Control




### HALLMARK TNFA SIGNALING VIA NFKB

N4 minus Control | NES 2.42
BH FDR 1.42e-13 | favours N4




Lowest-FDR pathway in each direction, regardless of significance. Pig orthologue projection; no claim of pathway activation.